

[Listing 8-1](#): A function that inserts spaces into texts, combining much of the code in the chapter so far

Then we can define any text and call our function as follows:

```
text = "The oneperfectly divine thing, the oneglimpse of God's paradisegiven on  
earth, is to \fight a losingbattle - and notlose it."  
print(insertspaces(text,word_list))
```

We see the output just as we expect, with spaces inserted perfectly:

```
The one perfectly divine thing, the one glimpse of God's paradise given on earth,  
is to fight a losing battle - and not lose it.
```

We’ve created an algorithm that can correctly insert spaces into English text. One thing to consider is whether you can do the same for other languages. You can—as long as you read in a good, appropriate corpus for the language you’re working with to define the `word_list`, the function we defined and called in this example can correctly insert spaces into text in any language. It can even correct a text in a language you’ve never studied or even heard of. Try different corpuses, different languages, and different texts to see what kind of results you can get, and you’ll get a glimpse of the power of language algorithms.

Phrase Completion

Imagine that you are doing algorithm consulting work for a startup that is trying to add features to a search engine they are building. They want to add phrase completion so that they can provide search suggestions to users. For example, when a user types in `peanutbutter` and, a search suggestion feature might suggest adding the word `jelly`. When a user types in `squash`, the search engine could suggest both `court` and `soup`.

Building this feature is simple. We’ll start with a corpus, just like we did with our space checker. In this case, we’re interested not only in the individual words of our corpus but also in how the words fit together, so we’ll compile lists of n -grams from our corpus. An n -gram is simply a collection of n words that appear together. For example, the phrase “Reality is not always probable, or likely” is made up of seven words once spoken